

ATMOS E3.S AI Capacity Portfolio

Three enterprise deployment modes. One separate hyperscale neural-data track.

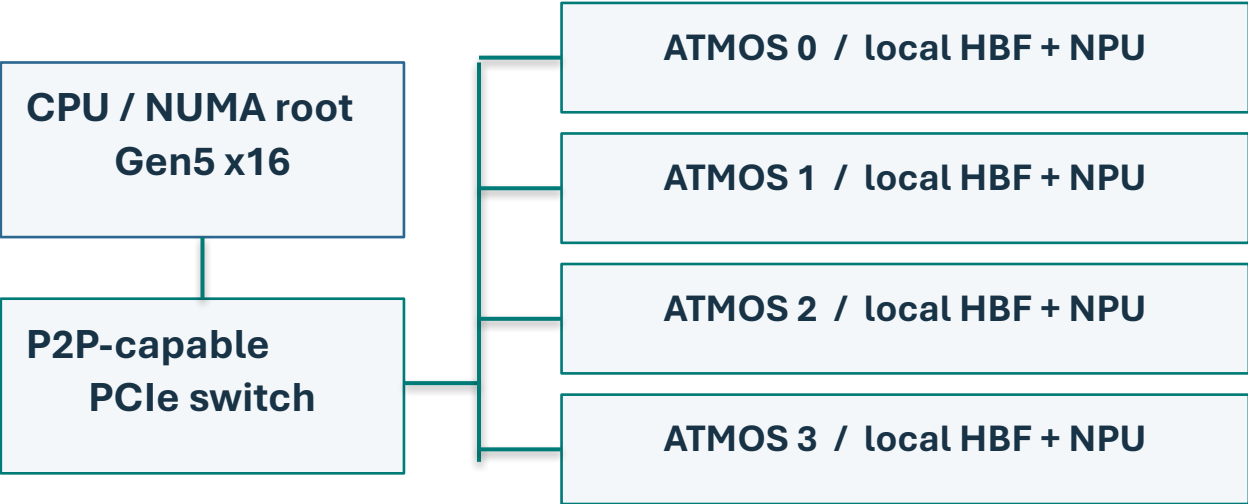
E1	ATMOS AI Appliance	Enterprise / private neural services
E2	ATMOS Companion Service	Existing enterprise GPU estates
E3	ATMOS Integrated Capacity Pool	Enterprises with platform operations teams
H	Hyperscale Neural-Data Sidecar	Hyperscalers and very large search / recommendation operators

LEAD E2: prove one complete service. Productize E1; integrate E3. Sponsor H only at scale.

Localize bytes. Exchange compact results.

One qualified four-device group is the atomic hardware and scheduling unit, not shared memory.

FOUR-DEVICE SWITCH / NUMA DOMAIN



STACK: REUSE UPSTREAM; QUALIFY THE ATMOS PATH

- Gateway / identity / quota / complete-request routing
- KServe or llm-d / model API and live endpoint metrics
- DRA / CDI + ATMOS Operator / ready-drain-rollback
- ATMOS workload runtime / selected NPU SDK / packs**
- ATMOS driver, firmware, DMA, P2P and telemetry

128 / 100 GB

Nominal HBF / study budget per device

200 GB/s

Local HBF read target; not peer bandwidth

~15.75 GB/s

Gen5 x4 / direction before overhead

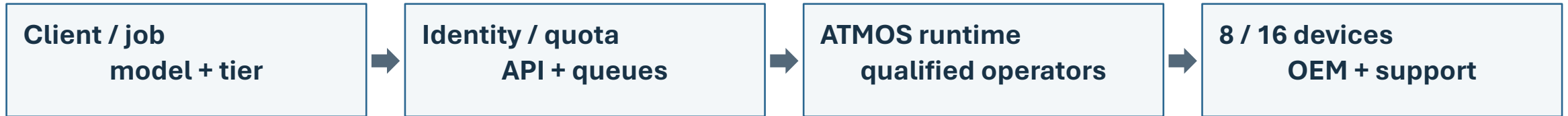
8 -> 16

Qualified node densities; 20-22 stretch

Host staging first. No unsafe isolation bypass. P2P/TT plugin/DRA is not ATMOS-ready model support. [S1-S3, S9]

E1 / A turnkey private neural-service endpoint

Sell a complete, supported workload first; full interactive LLM support is a separate qualification.



PRODUCT AND RELEASE ENVELOPE

- Lead with embeddings, rerank, retrieval or bounded document/batch operators.
- 8 devices: 1.024 TB nominal HBF. 16: 2.048 TB, distributed across qualified domains.
- Turnkey means identity, tenant isolation, packs, canary/drain, metering, HA and one support prime.

ILLUSTRATIVE 3-YEAR TCO SAVING

51.6%

**10.8 months
simple payback**

\$120,000 upfront | \$47,208/year ATMOS

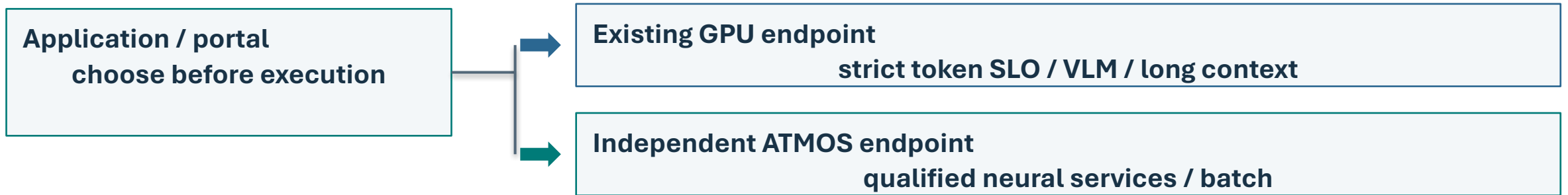
Cash benefit needs >48.4% baseline-spend realization.

SCENARIO: replace \$180k/year private service; \$120k upfront; 3 x 8-device nodes including N+1.

Compare with CPU / managed API too. Positive ROI is conditional on the best permitted baseline. [S13]

E2 / Add capacity without splitting an inference request

First enterprise integration priority: separate endpoint, failure domain and device lifecycle.



LOW-INTEGRATION ADOPTION / NON-NEGOTIABLE TESTS

- No CUDA/ROCm, GPU firmware or native-collective modification. No peer-DMA dependency.
- Shared gateway, CPU, NIC, storage and power can still hurt GPU tails: test mixed saturation.
- Freed installed GPU capacity is not cash. Avoid a specific future purchase or bill.

ILLUSTRATIVE 3-YEAR TCO SAVING

35.8%

**15.7 months
simple payback**

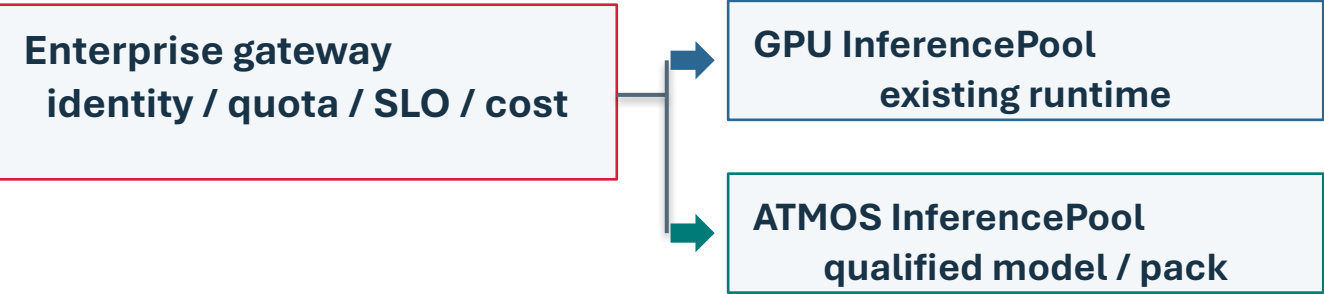
\$100,000 upfront | \$43,766/year ATMOS

Cash benefit needs >64.2% baseline-spend realization.

SCENARIO: avoid \$120k/year GPU overflow; \$100k upfront; 3 x 8-device nodes, including N+1.

E3 / One gateway. Two qualified hardware pools.

Reuse the enterprise catalog and policy; select one compatible, ready backend for the whole request.



DRA ALLOCATES DEVICES; THE OPERATOR MAKES A SERVICE READY

Contract	ATMOS responsibility
DRA / CDI	Group, NUMA, topology, allocation and isolation
Operator	ABI / pack / warmup -> ready -> drain / rollback
Gateway	Same model/API quality; live queue and safe overload

ILLUSTRATIVE 3-YEAR TCO SAVING

15.7% 10.0 months
simple payback

\$180,000 upfront | \$82,943/year ATMOS

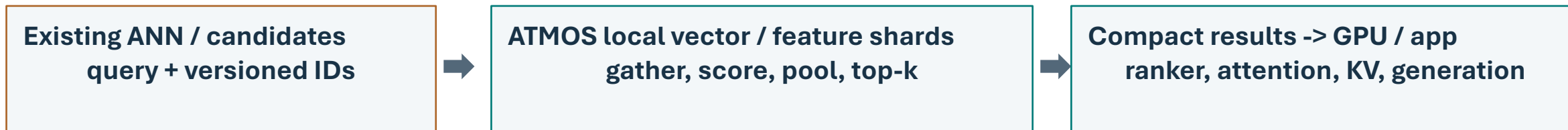
Cash benefit needs >47.6% baseline-spend realization.

vs E2: unified platform, identity and chargeback; more lifecycle/routing integration.

SCENARIO: \$1m/year estate; \$300k removable, \$700k retained; \$180k upfront. Full-estate denominator.

H / Hyperscale Neural-Data Sidecar

Hyperscalers and Meta / LinkedIn-like operators; enterprise only with exceptional scale and proven pain.



MEMORY AND LIFECYCLE / HYPOTHETICAL

500m x 768 x FP16
= 768 GB raw vectors

960 GB with a 25% layout allowance.
Four 16-device nodes: 2 generations x 2 complete replicas; each shard must fit.

REDUCE BEFORE EXPENSIVE RANKING

20,000 -> 200

99% fewer candidates, only if relevance is retained; not 99% total savings.

30.72 MB vectors -> 2.4 kB IDs/scores
ONLY for a vector-moving baseline. Input IDs / refresh / features excluded.

ILLUSTRATIVE WHOLE-PIPELINE TCO

19.4%

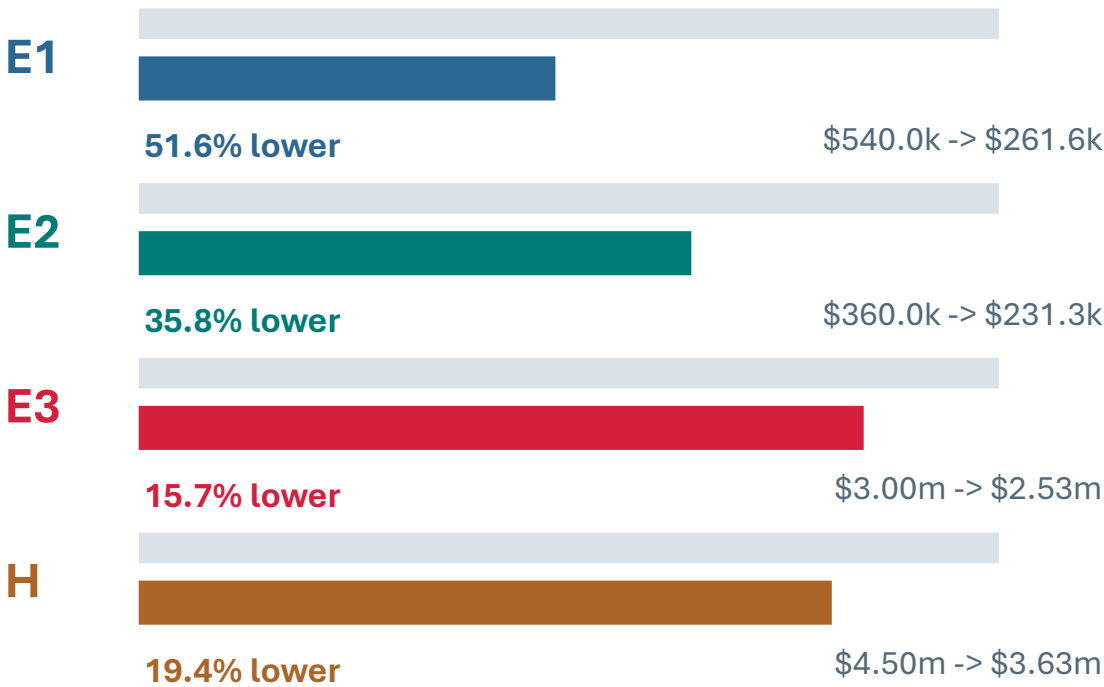
\$600k/year tier removable;
\$900k/year GPU retained. \$360k upfront, including replicas / staging nodes.

Gate: best CPU/DRAM ANN control + recall/NDCG + freshness + end-to-end p99 + deletions/ACL + actual tier removal. [S4, S10-S11]

Cost advantage needs accepted service AND avoided spend

Illustrative constant-USD 36-month cash models. Same quality, SLO, demand and availability; not ATMOS quotes or benchmarks.

3-YEAR TCO / EACH BASELINE NORMALIZED TO 100



100% OF SPECIFIED AVOIDABLE BUDGET REALIZED

Track	Upfront	3y net benefit	Payback	3y ROI
E1	\$120.0k	\$278,377	10.8 mo	232%
E2	\$100.0k	\$128,702	15.7 mo	129%
E3	\$180.0k	\$471,170	10.0 mo	262%
H	\$360.0k	\$873,510	10.5 mo	243%

E3 retains \$700k/year GPU services. H retains \$900k/year. Denominators cover the whole unchanged service/estate.

Includes hardware, rollout, support, software, operations, electricity and reserve. No double-counting / no additive portfolio savings.

Where the business case breaks

Realized fraction means baseline bills or future purchases actually avoided, not just requests routed elsewhere.

NET 3-YEAR BENEFIT AT DIFFERENT REALIZATION LEVELS

Track	50% realized	75% realized	100% realized	Break-even
E1	\$8,377	\$143,377	\$278,377	>48.4%
E2	-\$51,298	\$38,702	\$128,702	>64.2%
E3	\$21,170	\$246,170	\$471,170	>47.6%
H	-\$26,490	\$423,510	\$873,510	>51.5%

- Freed GPU capacity is noncash until a bill or future purchase is avoided.
- Stress-test goodput, node count, refresh/HA, support/NRE and energy.

THE CHEAPER ALTERNATIVE CAN WIN

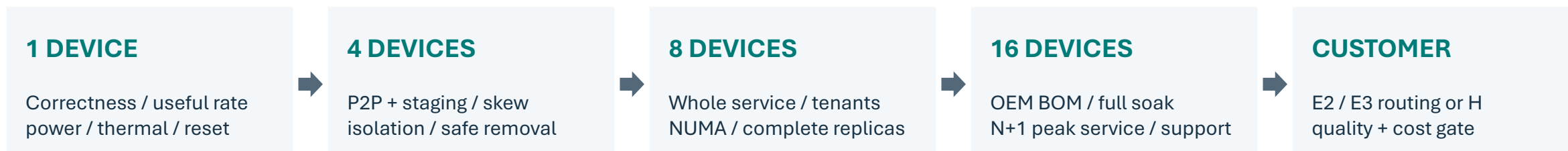
\$1,024/year

Published small-embedding API token charges at 100m x 512 tokens. \$0.02 per million tokens. [S13]

API is not an equal private-service quote. If privacy, quality and SLO permit it, the assumed appliance baseline must change.

OEM density and service are evidence gates, not bay counts

A four-device fixture -> supported 8-device pilot -> 16-device standard. Hold 20-22 until customer/OEM proof.



WHAT THE OEM / PRIME MUST SIGN

- E3.S 2T / x4 lanes / BAR and security / BIOS-BMC / sustained bay power and cooling.
- Wall power at maximum inlet; CPU/NIC/storage load; safe replacement and full replica reserve.
- Named BOM, NRE, firmware/driver/runtime matrix, spares and one support escalation route.

WHAT THE SERVICE / FINANCE GATE MUST PROVE

- Same quality, p99 / TTFT / TPOT where relevant, freshness and accepted service under failures.
- Protect GPU tails and tenant isolation; no forced unqualified routing or unsafe DMA bypass.
- Best incumbent/CPU/GPU comparison + measured node count + actual quotes + removable spend.

Public 32-bay E3.S 1T x2 storage density does not certify ATMOS x4/2T accelerators. Simulation cannot pass physical gates. [S1, S6]

Fund bounded proof. Hold launch and savings claims.

One shared foundation, clear ownership, four retained tracks and a linked technical/economic decision package.

NEXT DECISIONS / ACCOUNTABLE OWNERS

E2 FIRST	Product + customer: freeze one paid workload, best control and removable overflow block.
E1 PRODUCT	OEM + SanDisk: qualify 4 -> 8 -> 16 devices, full-service reserve and one support prime.
E3 INTEGRATE	Platform owner: DRA/operator/gateway contract, GPU tail protection and cost-aware policy.
H AT SCALE	Hyperscale sponsor: index/update trace, relevance/freshness, p99 and tier-removal charter.
FINANCE GATE	Finance + engineering: substitute matched physical rates and binding quotes; reject below break-even.

LINKED WORD DETAIL NAVIGATOR

2 Foundation / software boundary

3-5 Enterprise E1 / E2 / E3

6 Hyperscale H / sizing / refresh

8 Cost / ROI / downside / API control

9-11 OEM / support / gates

A Research / source and claim register

Keep PPT + DOCX together.
Use 'Open Word proposal' if a viewer ignores section bookmarks.